

Topic: Intersection Operation

Q. Describe the Intersection operation in Relational Algebra. What are the required conditions for applying it? Illustrate its use with a suitable example.

📌 Definition to Remember

The **Intersection Operation** ($\cap$) is a **binary set operation** in relational algebra that returns only those tuples (rows) that are **common to both** input relations. It is a **derived operator** (can be expressed as $A \cap B = A - (A - B)$) and requires both relations to be **Union Compatible**.

1. Intersection Operator ($\cap$)

- **Binary operation** — operates on two tables.
- **Syntax:** Relation_A $\cap$ Relation_B
- Returns only tuples that appear in **both** relations simultaneously.
- It is a **derived operator**: $A \cap B = A - (A - B)$
- Equivalent to `INTERSECT` in SQL.

2. Required Conditions (Union Compatibility)

Both relations must be **Union Compatible**:

1. **Same Degree:** Same number of columns.
2. **Same Domain:** Corresponding columns have matching data types.

3. Example

TOP_IN_MATH

Roll_No	Name	Grade
101	John	A
102	Emma	A
104	Mia	B

TOP_IN_SCIENCE

Roll_No	Name	Grade
102	Emma	A
105	Noah	A
101	John	B

Query: Find students with **identical records** (same Roll_No, Name, and Grade) in both tables.

`$TOP_IN_MATH  $\cap$  TOP_IN_SCIENCE$`

Result: Only **Emma (102, A)** — her entire tuple (Roll_No=102, Name=Emma, Grade=A) is identical in both tables.

(Note: John appears in both tables but with different grades — A in Math, B in Science — so his full tuple is NOT identical, and he is excluded.)

★ **Must-Write Points (for 10 marks)**

1. Intersection ($\cap$) returns only rows that are **identical in both** relations.
2. It is a **derived operator**: $A \cap B = A - (A - B)$.
3. **Union Compatibility** required: same number of columns AND matching data types.
4. The **entire tuple** must be identical — not just one column — for a row to be included.
5. If $|A| = m$ and $|B| = n$, result size $\leq \min(m, n)$.
6. Equivalent to SQL `INTERSECT`.
7. Used to find commonalities between two datasets (e.g., students excelling in both Math and Science).

⚡ **Quick Recall**

$\cap$ (Intersection) → Binary → Returns Common Rows Only → Derived from Set Difference → Union Compatibility → SQL `INTERSECT`